

**YEAR 10 SCIENCE EXTENSION
CHEMISTRY
TEST 2: STOICHIOMETRY**

Name: SOLUTIONS Form: _____ Mark: 25

1. Give the formula and formula mass for the following compounds.

NAME	FORMULA	FORMULA MASS (g mol^{-1})
lead (II) nitrate	$\text{Pb}(\text{NO}_3)_2$	331.22
potassium hydrogencarbonate	KHCO_3	100.12
calcium phosphate	$\text{Ca}_3(\text{PO}_4)_2$	310.18
aluminium chloride	AlCl_3	133.33

↑ (1 mark each)

↑ (1 mark each)

(8)

2. How many moles are in 3.25 g of $\text{Mg}(\text{NO}_3)_2$?

$$\begin{aligned}
 n(\text{Mg}(\text{NO}_3)_2) &= \frac{m}{M} & M(\text{Mg}(\text{NO}_3)_2) &= 148.33 \text{ g mol}^{-1} \quad (1) \\
 &= \frac{3.25}{148.33} \quad (1) \\
 &= \underline{2.19 \times 10^{-2} \text{ mol}} \quad (1)
 \end{aligned}$$

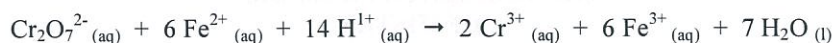
(3)

3. Calculate the mass of 5.46 moles of $\text{Zn}(\text{CH}_3\text{COO})_2$.

$$\begin{aligned}
 m(\text{Zn}(\text{CH}_3\text{COO})_2) &= n \times M & M(\text{Zn}(\text{CH}_3\text{COO})_2) &= 183.47 \text{ g mol}^{-1} \quad (1) \\
 &= (5.46)(183.47) \quad (1) \\
 &= \underline{1.00 \times 10^3 \text{ g}} \quad (1)
 \end{aligned}$$

(3)

4. The following is a complex ionic equation for a particular oxidation/reduction reaction.



Complete the following mole ratios for this reaction.

$$n(\text{Cr}_2\text{O}_7^{2-}) = \underline{\frac{1}{6}} \times n(\text{Fe}^{3+})$$

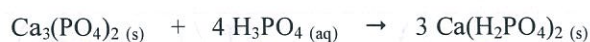
$$n(\text{H}^{+}) = \underline{2} \times n(\text{H}_2\text{O})$$

$$n(\text{Cr}^{3+}) = \underline{\frac{1}{3}} \times n(\text{Fe}^{2+})$$

(1 mark each)

(3)

5. The overall reaction for the manufacture of superphosphate is given below. 25.0 tonnes (2.5×10^4 kg) of calcium phosphate is reacted.



- (a) Calculate the mass of superphosphate ($\text{Ca}(\text{H}_2\text{PO}_4)_2$) produced?

$$n(\text{Ca}_3(\text{PO}_4)_2) = \frac{m}{M}$$

$$M(\text{Ca}_3(\text{PO}_4)_2) = 310.18 \text{ g mol}^{-1}$$

$$= \frac{2.50 \times 10^7}{310.18} \quad (1\frac{1}{2})$$

$$= 8.06 \times 10^4 \text{ mol} \quad (1\frac{1}{2})$$

$$n(\text{Ca}(\text{H}_2\text{PO}_4)_2) = \frac{3}{1} \times n(\text{Ca}_3(\text{PO}_4)_2) \quad (1)$$

$$= 3(8.06 \times 10^4)$$

$$= 2.418 \times 10^5 \text{ mol} \quad (1\frac{1}{2})$$

$$m(\text{Ca}(\text{H}_2\text{PO}_4)_2) = n \times M$$

$$M(\text{Ca}(\text{H}_2\text{PO}_4)_2) = 203.08 \text{ g mol}^{-1}$$

$$= (2.418 \times 10^5)(203.08) \quad (1)$$

$$= 4.91 \times 10^7 \text{ g} \quad (1\frac{1}{2})$$

(5)

(b) Determine how much phosphoric acid (H_3PO_4) is needed.

$$n(\text{H}_3\text{PO}_4) = \frac{4}{1} \times n(\text{Ca}_3(\text{PO}_4)_2) \quad (1)$$

$$= 4(8.06 \times 10^4)$$

$$= 3.224 \times 10^5 \text{ mol} \quad \left(\frac{1}{2}\right)$$

$$m(\text{H}_3\text{PO}_4) = n \times M \quad M(\text{H}_3\text{PO}_4) = 97.994 \text{ g mol}^{-1}$$

$$= (3.224 \times 10^5)(97.994) \quad (1)$$

$$= \underline{3.16 \times 10^7 \text{ g}} \quad \left(\frac{1}{2}\right)$$

(3)

